

1. Convexity. Is the function $f(x_1, x_2) = (x_1 + x_2)^2$ convex?

$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \implies \lambda \in \{0, 2\} \implies f \text{ convex}$$

2. Strict Convexity: Is the function $f(x_1, x_2) = (x_1 + x_2)^2$ strictly convex?

$\lambda = 0$ so not strictly convex.

3. Optimization: A convex region is defined similarly to a convex function:

- pick two points
- draw a line between them
- if all the points on the line are contained in the region, the region is convex
- if not, then it's not convex

a. Please show that when minimizing a strictly convex function $f(\cdot)$ over its entire domain that the minimizing point is unique.

Suppose $f(x) = f(y) = \min f(\cdot)$ but $x \neq y$. Then

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y) = f(y)$$

but this contradicts minimality of y .

b. Please show that the minimum of a strictly convex $f(x)$ on a non-convex region might not be unique.

c. If a continuous function $f(x)$ is strictly convex over $x \in \mathbb{R}^n$, then does the absolute minimum occur at the point where all the N first partials are zero?

Yes.

d. If a function is NOT convex, can we be assured the minimum occurs when the first partials are zero?

No. Consider $f(x) = x^3(x - 2)$. Not convex and global minimum is at $x = 2$ but $f_x(0) = 4x^2 - 6x^2 = 0$.

4. Quantization and Convexity: You are designing a 2-level quantizer with parameters x_0, q_0, q_1 .

a. Please derive the Hessian matrix for the three design parameters.

$$Q(x) = \begin{cases} q_0 & x < x_0 \\ q_1 & x \geq x_0 \end{cases} = q_0 \mathbb{1}_{x \in [-\infty, x_0)}(x) + q_1 \mathbb{1}_{x \in [x_0, \infty)}(x)$$

$$H_Q = (1)$$

b. Is the Hessian matrix positive definite? Positive semi-definite?